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1 General Information

1.1 Academic Qualifications

- **Ph.D.** in Electrical Engineering and Computer Science, 2008
  Massachusetts Institute of Technology, USA
  Topic: Decentralized Detection in Resource-limited Sensor Network Architectures
  Advisors: John Tsitsiklis and Moe Z. Win
- **MS** in Electrical Engineering, 2002
  Stanford University, USA
- **BS** in Mathematics and Electrical Engineering with *distinction*, 2002
  Stanford University, USA

1.2 Awards and Honors

- **Outstanding Editorial Board Member Award**, 2023.
  IEEE Transactions on Signal and Information Processing Over Networks.
- **Excellent Paper Award**, International Conference on Smart Power & Internet Energy Systems, 2022.
- **IEEE Signal Processing Society Young Author Best Paper Award**, 2016.
  Awarded with Jack Ho. I am a co-author of the paper and the main supervisor of Jack.
- **Early Career Teaching Excellence Award**, 2016.
  School of Electrical and Electronic Engineering, Nanyang Technological University. In recognition of tenure-track assistant professors who have consistently demonstrated excellent teaching.
- **Tan Chin Tuan Exchange Fellowship in Engineering**, 2015.
- **Best Student Paper Award**, 46th Asilomar Conference on Signals, Systems, and Computers, 2012.
  The Asilomar Conference on Signals, Systems, and Computers is a prestigious conference in signal processing and computer systems, and is technically co-sponsored by the IEEE Signal Processing Society.
- **Frederick Emmons Terman Engineering Scholastic Award, Stanford University**, 2002.
  Awarded to the top engineering students.
- **President's Award, Stanford University**, 1999.
  Awarded to top students for academic excellence.

1.3 Work Experience

- Sep. 2023 – present, Professor of Signal and Information Processing, School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore
- Sep. 2017 – Aug. 2023, Associate Professor (with tenure), School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore
- Mar. 2010 – Aug. 2017, Assistant Professor, School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore
- Oct. 2008 – Mar. 2010, Associate, Quantitative Strategy, Morgan Stanley Asia Limited, Hong Kong
- Feb. 2008 – Sep. 2008, Associate, Quantitative Credit Research, Lehman Brothers Asia Limited
- Sep. 2004 – Dec. 2007, Research Assistant/Teaching Assistant, Massachusetts Institute of Technology
- Aug. 2002 – Jul. 2004, Various industry/academic positions due to scholarship bond, Singapore/US

1.4 Consulting Experience

- Apr. 2010 – Mar. 2012, Financial Modelling, Morgan Stanley Asia Limited

1.5 Professional Memberships

- Senior Member, IEEE

2 Research Activities

2.1 Ongoing Research Grants

- PI, "PERCEPDRONE", Ministry of Defence, Sep. 2025 – Mar. 2029, S$1,245,640
- PI, "Project Nemo: UE Localization and Tracking", DSO National Laboratories, Sep. 2025 – Sep. 2026, S$150,000
- PI, "Project Navigator: Aerial-ground Place Recognition with Pose Regression", National University of Singapore, May 2025 – May 2026, S$146,300
- PI, "Project Trio", DSO National Laboratories, May 2025 – May 2026, S$172,450
- PI, "Project Spyder", DSO National Laboratories, May 2025 – May 2026, S$335,000

2.2 Completed Research Grants

- PI, "Generalized Message Passing for Federated Learning in Multi-Access Edge Computing", Future Communications Program, Jun. 2022 – May 2025, S$461,538
- PI, "Project Ananke 2: Aerial-Ground Place Recognition for Drones", DSO National Laboratories, Jul. 2024 – Dec. 2024, S$50,000
- PI, "Project Ananke: Robust Multi-Modal Neural Diffusion for Light-Weight Drone Perception", DSO National Laboratories, Jul. 2023 – Jun. 2024, S$100,000
- PI, "Project LOCUS: RF Localization Using Donor Signals", DSO National Laboratories, Sep. 2022 – Nov. 2024, S$499,999.50
- PI, "Graph Signal Processing for High Dimensional Structures and Spaces", Ministry of Education Academic Research Fund Tier 2, Nov. 2021 – Oct. 2024, S$567,972
- PI, "Research and Applications of 5G-V2X Communication and Computing", NI Singapore Ltd, Oct. 2020 – Oct. 2022, S$120,000
- PI, "5G-V2X Communication Trial and Use Cases Development", M1 Limited, Jan. 2020 – Dec. 2023, S$98,040
- PI, "End to End Learned Localization for AV", NRF/Continental Corp Lab/NTU, Dec. 2019 – Nov. 2022, S$595,909
- PI, "Next-Generation V2X Network Architecture and Ecosystem for Smart Mobility", A*Star Industry Alignment Fund - Pre Positioning (IAF-PP), Nov. 2019 – Oct. 2023, S$21,240,000
- Co-PI, "Design and Reinforcement Security on Smart Grids Against Cyber-Physical Attack", NRF, Oct. 2019 – Mar. 2023, S$271,700 (total grant: S$997,460)
- PI, "Decentralized Privacy for the Internet of Things: Theory and Algorithms", Ministry of Education Academic Research Fund Tier 2, May 2019 – Jul. 2022, S$568,512

- Co-PI, "Outdoor Geo-Localization and Navigation Using DVB and LTE Signals-of-Opportunity in GPS-denied Environments", Ministry of Defence, Aug. 2018 – Aug. 2020, S$300,000 (total grant: S$1,200,000)
- PI, "Relative Positions Determination Using Communication Signals", Defence Science Organization National Laboratories, Aug. 2018 – Apr. 2020, S$479,159
- PI, "Energy Efficient Resilient Distributed Inference", NRF/Delta Electronics Corp Lab/NTU, Jul. 2016 – Aug. 2020, S$434,600
- PI, "Fake News Detection: A Graph Signal Processing And Learning Approach", Ministry of Education Academic Research Fund Tier 1, Nov. 2017 – Jul. 2019, S$80,000
- Co-PI, "Project Moscato: A Holistic Approach to Combatting Insider Threats", Ministry of Defence, Dec. 2015 – Jun. 2019, S$308,000 (total grant: S$4,762,600)
- Co-PI, "UWB Based Collaborative Decentralised Localisation", NRF/ST Dynamics Corp Lab/NTU, Jul. 2015 – Jun. 2019, S$427,400 (total grant: S$1,187,000)
- Co-PI, "Feasibility Study of Multi-Function Millimeter-Wave RaCoPo System", Huawei International Pte. Ltd., Oct. 2017 – Apr. 2019, S$148,944 (total grant: S$417,600)
- PI, "Multi-Target Track Before Detect On TDOA-FDOA Based Measurements", Defence Science Organization National Laboratories, Mar. 2015 – Feb. 2016, S$100,000
- PI, "Robust Learning in Social Networks: Fundamental Limits and Strategies", Ministry of Education Academic Research Fund Tier 2, Jan. 2015 – Jun. 2018, S$652,998
- Co-PI, "Development of NTU/NXP-Intelligent Transport System Test-Bed", Economic Development Board, Dec. 2014 – Nov. 2019, S$2,568,300 (total grant: S$14,098,000)
- PI, "Identifying Infection Sources in a Network", Ministry of Education Academic Research Fund Tier 2, Apr. 2014 – Mar. 2017, S$507,593
- Co-PI, "Project Urban-Nav: Urban Outdoor Navigation of Unmanned Platform under a GPS Challenged Environment", Defence Research and Technology Office, Dec. 2013 – Feb. 2017, S$223,000 (total grant: S$1,854,000)
- PI, "GPS Free TDOA/FDOA Geolocation and Tracking II", Defence Science Organization National Laboratories, Mar. 2013 – Nov. 2015, S$236,500
- PI, "Project Technifibre: Cooperative and Distributed Tracking in Urban Environments", Defence Science and Technology Agency, May 2012 – May 2015, S$622,725
- PI, "Intelligent Information Fusion and Inference in Sensor Networks", Ministry of Education Academic Research Fund Tier 1, Mar. 2011 – Feb. 2014, S$100,000
- PI, "UWB Monitoring System for Sleep Studies", Nanyang Institute of Technology in Health and Medicine, Jul. 2011 – Jan. 2013, S$70,000
- PI, "GPS Free TDOA/FDOA Geolocation and Tracking", Defence Research and Technology Office, Nov. 2011 – Nov. 2012, S$65,200
- PI, "Distributed Signal Processing and Algorithms for Decentralized Decision Making and Computation in Sensor Networks", NTU Startup Grant, Mar. 2010 – Mar. 2013, S$100,000

2.3 Published/Accepted Journal Papers

J1. T. Wei, K. V. Mishra, M. Bhavani Shankar, W. P. Tay, and B. Ottersten, "Fundamental limits for near-field sensing — Part II: Wide-band systems," *IEEE Transactions on Antennas and Propagation*, in press

J2. T. Wei, K. V. Mishra, M. Bhavani Shankar, W. P. Tay, and B. Ottersten, "Fundamental limits for near-field sensing — Part I: Narrow-band systems," *IEEE Transactions on Antennas and Propagation*, in press

J3. X. Tang, Y. Yan, W. P. Tay, and X. Cao, "A privacy-preserving scheme with system-dimension masking for cloud-based model predictive control," *Automatica*, vol. 192, p. 113152, 2026

J4. R. She, S. Wang, Q. Kang, K. Zhao, T. Geng, Y. Zhao, W. Liang, and W. P. Tay, "Hierarchical information embeddings with neural ODEs for personalized federated learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 48, no. 7, pp. 7241 – 7257, 2026

J5. X. Liu, Y. Yuan, Z. Zhang, Z. Ma, W. P. Tay, and P. Fan, "Customized OTFS pulse compression and waveform optimization for enhancing target sensing in ISAC systems," *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 760 – 775, 2026

J6. C. X. Wang and W. P. Tay, "Data-driven regularized inference privacy," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 6, pp. 6862 – 6877, 2025

J7. X. Jian, M. Golz, F. Ji, W. P. Tay, and A. M. Zoubir, "A graph signal processing perspective of network multiple hypothesis testing with false discovery rate control," *IEEE Transactions on Signal Processing*, vol. 73, pp. 3496 – 3512, 2025

J8. F. Ji, Y. Zhao, S. H. Lee, K. Zhao, W. P. Tay, and J. Yang, "Graph distributional signals for regularization in graph neural networks," *IEEE Transactions on Signal and Information Processing over Networks*, vol. 11, pp. 670 – 682, 2025

J9. X. Ran, W. P. Tay, and C. H. T. Lee, "High-performance optimization model based on novel conditional value at risk metric for power grids with high wind power penetration," *IEEE Transactions on Industrial Informatics*, vol. 21, no. 7, pp. 5689 – 5700, Jul. 2025

J10. X. Ran, W. P. Tay, and C. H. T. Lee, "Uncertain interval-based risk dispatch approach of power systems under an unified framework of multiple uncertainties," *IEEE Transactions on Industrial Informatics*, vol. 21, no. 3, pp. 2053 – 2063, Mar. 2025

J11. Y. Yuan, W. Li, Y. Liang, W. P. Tay, W. Gao, and F. Shen, "Lane-aware dynamic spatio-temporal transformer for unimodal trajectory prediction," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 2, pp. 1156 – 1167, Feb. 2025

J12. V. Mohan, W. P. Tay, and A. Basu, "Towards neuromorphic compression based neural sensing for next-generation wireless implantable brain machine interface," *Neuromorphic Computing and Engineering*, vol. 5, no. 1, p. 014004, Jan. 2025

J13. S. H. Lee, F. Ji, and W. P. Tay, "Node-specific space selection via localized geometric hyperbolicity in graph neural networks," *Transactions on Machine Learning Research*, 2024. [Online]. Available: `https://openreview.net/forum?id=tHteJFeN1y`

J14. S. Wang, Q. Kang, R. She, K. Zhao, Y. Song, and W. P. Tay, "PRFusion: Toward effective and robust multi-modal place recognition with image and point cloud fusion," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 12, pp. 20 523 – 20 534, 2024

J15. F. Ji, X. Jian, and W. P. Tay, "Modeling sparse graph sequences and signals using generalized graphons," *IEEE Transactions on Signal Processing*, vol. 72, pp. 5048 – 5064, 2024

J16. Y. Lu, F. Ji, and W. P. Tay, "Heterogeneous graph adaptive flow network," *Transactions on Machine Learning Research*, 2024. [Online]. Available: `https://openreview.net/forum?id=usvg3yhjAx`

J17. X. Ran, W. P. Tay, and C. H. T. Lee, "Robust data-driven adversarial false data injection attack detection method with deep Q-network in power systems," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 8, pp. 10 405 – 10 418, 2024

J18. X. Liu, Y. Yuan, T. Zhang, G. Cui, and W. P. Tay, "Integrated transmit waveform and RIS phase shift design for LPI detection and communication," *IEEE Transactions on Wireless Communications*, vol. 23, no. 6, pp. 5663 – 5679, Jun. 2024

J19. X. Jian, W. P. Tay, and Y. Eldar, "Kernel based reconstruction for generalized graph signal processing," *IEEE Transactions on Signal Processing*, vol. 72, pp. 2308 – 2322, 2024

J20. A. Darendeli, A. Sun, and W. P. Tay, "The geography of corporate fake news," *PLOS ONE*, vol. 19, no. 4, pp. 1 – 28, Apr. 2024

J21. Q. Kang, W. P. Tay, R. She, S. Wang, X. Liu, and Y.-R. Yang, "Multi-armed linear bandits with latent biases," *Information Sciences*, vol. 660, p. 120103, 2024

J22. R. She, Q. Kang, S. Wang, W. P. Tay, K. Zhao, Y. Song, T. Geng, Y. Xu, D. N. Navarro, and A. Hartmannsgruber, "PointDifformer: Robust point cloud registration with neural diffusion and transformer," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1 – 15, 2024

J23. R. She, Q. Kang, S. Wang, Y.-R. Yang, K. Zhao, Y. Song, and W. P. Tay, "RobustMat: Neural diffusion for street landmark patch matching under challenging environments," *IEEE Transactions on Image Processing*, vol. 32, pp. 5550 – 5563, 2023

J24. R. She, Q. Kang, S. Wang, W. P. Tay, Y. L. Guan, D. N. Navarro, and A. Hartmannsgruber, "Image patch-matching with graph-based learning in street scenes," *IEEE Transactions on Image Processing*, vol. 32, pp. 3465 – 3480, 2023

J25. F. Ji, W. P. Tay, and A. Ortega, "Graph signal processing over a probability space of shift operators," *IEEE Transactions on Signal Processing*, vol. 71, pp. 1159 – 1174, 2023

J26. W. Zhang, Z. Wang, and W. P. Tay, "Approximate maximum-likelihood RIS-aided positioning," *IEEE Transactions on Wireless Communications*, vol. 22, no. 12, pp. 8859 – 8875, Dec. 2023

J27. H. Cheng, J. T. Zhou, W. P. Tay, and B. Wen, "Graph neural networks with triple attention for few-shot learning," *IEEE Transactions on Multimedia*, vol. 25, pp. 8225 – 8239, 2023

J28. X. Liu, Y. Yuan, T. Zhang, G. Cui, and W. P. Tay, "LPI radar signal design resistant to identification by ESM systems," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 6, pp. 9233 – 9246, Dec. 2023

J29. P. Lin, C. Deng, Y. Yang, C. H. T. Lee, and W. P. Tay, "Resilience-oriented control for cyber-physical hybrid energy storage systems using a semi-consensus scheme: Design and practice," *IEEE Transactions on Industrial Electronics*, vol. 70, no. 3, pp. 2508 – 2519, Mar. 2023

J30. X. Jian and W. P. Tay, "Wide-sense stationarity in generalized graph signal processing," *IEEE Transactions on Signal Processing*, vol. 70, pp. 3414 – 3428, 2022

J31. Y. Yan, I. Bajaj, R. Rabiee, and W. P. Tay, "A tightly coupled integration approach for cooperative positioning enhancement in DSRC vehicular networks," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 12, pp. 23 278 – 23 294, Dec. 2022

J32. J. Yang and W. P. Tay, "An unsupervised Bayesian neural network for truth discovery in social networks," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 11, pp. 5182 – 5195, Nov. 2022

J33. A. B. Bhutto, X. S. Vu, E. Elmroth, W. P. Tay, and M. Bhuyan, "Reinforced transformer learning for VSI-DDoS detection in edge clouds," *IEEE Access*, vol. 10, pp. 94 677–94 690, 2022

J34. Q. Kang and W. P. Tay, "Task recommendation in crowdsourcing based on learning preferences and reliabilities," *IEEE Transactions on Services Computing*, vol. 15, no. 4, pp. 1785–1798, 2022

J35. F. Ji, Pratibha, and W. P. Tay, "Unlimited dynamic range signal recovery for folded graph signals," *Signal Processing*, vol. 198, p. 108574, 2022

J36. F. Ji, G. Kahn, and W. P. Tay, "Signal processing on simplicial complexes with vertex signals," *IEEE Access*, vol. 10, pp. 41 889 – 41 901, 2022

J37. C. Wang, W. P. Tay, Y. Wei, and Y. Wang, "Privacy-preserving distributed projection LMS for linear multitask networks," *IEEE Transactions on Signal Processing*, vol. 69, pp. 6530 – 6545, 2021

J38. W. Zhang and W. P. Tay, "Cost-efficient RIS-aided channel estimation via rank-one matrix factorization," *IEEE Wireless Communications Letters*, vol. 10, no. 11, pp. 2562 – 2566, Nov. 2021

J39. T. S. Lau and W. P. Tay, "Asymptotically optimal sampling policy for quickest change detection with observation-switching cost," *IEEE Transactions on Signal Processing*, vol. 69, pp. 1332 – 1346, 2021

J40. C. X. Wang, Y. Song, and W. P. Tay, "Arbitrarily strong utility-privacy tradeoff in multi-agent systems," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 671 – 684, 2021

J41. Y. Song, C. X. Wang, and W. P. Tay, "Compressive privacy for a linear dynamical system," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 895 – 910, 2020

J42. M. Sun and W. P. Tay, "On the relationship between inference and data privacy in decentralized IoT networks," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 852 – 866, 2020

J43. M. Sun and W. P. Tay, "Decentralized detection with robust information privacy protection," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 85–99, 2020

J44. X. Liu, W. P. Tay, Z.-W. Liu, and G. Xiao, "Quasi-synchronization of heterogeneous networks with a generalized Markovian topology and event-triggered communication," *IEEE Transactions on Cybernetics*, vol. 50, no. 10, pp. 4200 – 4213, Oct. 2020

J45. F. Ji, W. Tang, W. P. Tay, and E. K. P. Chong, "Network topology inference using information cascades with limited statistical knowledge," *Information and Inference: A Journal of the IMA*, vol. 9, no. 2, pp. 327 — 360, Jun. 2020

J46. F. Ji and W. P. Tay, "A Hilbert space theory of generalized graph signal processing," *IEEE Transactions on Signal Processing*, vol. 67, no. 24, pp. 6188 – 6203, Dec. 2019

J47. X. He, W. P. Tay, H. Lei, M. Sun, and Y. Gong, "Privacy-aware sensor network via multilayer nonlinear processing," *IEEE Internet of Things Journal*, vol. 6, no. 6, pp. 10 834 – 10 845, Dec. 2019

J48. F. Wen, H. Wymeersch, B. Peng, W. P. Tay, H. C. So, and D. Yang, "A survey on 5G massive MIMO localization," *Digital Signal Processing*, vol. 94, pp. 21 – 28, Nov. 2019

J49. Q. Kang and W. P. Tay, "Sequential multi-class labeling in crowdsourcing," *IEEE Transactions on Knowledge and Data Engineering*, vol. 31, no. 11, pp. 2190 – 2199, Nov. 2019

J50. T. S. Lau and W. P. Tay, "Quickest change detection in the presence of a nuisance change," *IEEE Transactions on Signal Processing*, vol. 67, no. 20, pp. 5281 – 5296, Oct. 2019

J51. L. Ma, W. P. Tay, and G. Xiao, "Iterative expectation maximization for reliable social sensing with information flows," *Information Sciences*, vol. 501, pp. 621 – 634, Oct. 2019

J52. J. Yang, J. Wang, and W. P. Tay, "Using social network information in community-based Bayesian truth discovery," *IEEE Transactions on Signal and Information Processing over Networks*, vol. 5, no. 3, pp. 525 – 537, Sep. 2019

J53. R. Rabiee, X. Zhong, Y. Yan, and W. P. Tay, "LaIF: A lane-level self-positioning scheme for vehicles in GNSS-denied environments," *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, no. 8, pp. 2944 – 2961, Aug. 2019

J54. F. Ji, W. Tang, and W. P. Tay, "On the properties of Gromov matrices and their applications in network inference," *IEEE Transactions on Signal Processing*, vol. 67, no. 10, pp. 2624 – 2638, May 2019

J55. G. Yang, W. P. Tay, Y. L. Guan, and Y.-C. Liang, "Optimal power allocation for diffusion-type sensor networks with wireless information and power transfer," *IEEE Access*, vol. 7, pp. 32 408 – 32 422, Mar. 2019

J56. T. S. Lau, W. P. Tay, and V. V. Veeravalli, "A binning approach to quickest change detection with unknown post-change distribution," *IEEE Transactions on Signal Processing*, vol. 67, no. 3, pp. 609 – 621, Feb. 2019

J57. W. Tang, F. Ji, and W. P. Tay, "Estimating infection sources in networks using partial timestamps," *IEEE Transactions on Information Forensics and Security*, vol. 13, no. 2, pp. 3035 – 3049, Dec. 2018

J58. J. Yang, X. Zhong, and W. P. Tay, "A dynamic Bayesian nonparametric model for blind calibration of sensor networks," *IEEE Internet of Things Journal*, vol. 5, no. 5, pp. 3942 – 3953, Oct. 2018

J59. M. Z. A. Bhotto and W. P. Tay, "Non-Bayesian social learning with observation reuse and soft switching," *ACM Transactions on Sensor Networks*, vol. 14, no. 2, pp. 14:1–14:21, Jun. 2018

J60. M. Sun, W. P. Tay, and X. He, "Toward information privacy for the Internet of Things: A non-parametric learning approach," *IEEE Transactions on Signal Processing*, vol. 66, no. 7, pp. 1734 – 1747, Apr. 2018

J61. F. Quitin, P. De Doncker, F. Horlin, and W. P. Tay, "Virtual multi-antenna array for estimating the direction of a transmitter: system, bounds and experimental results," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 2, pp. 1510 – 1520, Feb. 2018

J62. Y. Yu, G. Xiao, G. Li, W. P. Tay, and H. F. Teoh, "Opinion diversity and community formation in adaptive networks," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 27, no. 20, p. 103115, Oct. 2017

J63. W. Luo, W. P. Tay, and M. Leng, "On the universality of Jordan centers for estimating infection sources in tree networks," *IEEE Transactions on Information Theory*, vol. 63, no. 7, pp. 4634 – 4657, Jul. 2017

J64. F. Ji, W. P. Tay, and L. Varshney, "An algorithmic framework for estimating rumor sources with different start times," *IEEE Transactions on Signal Processing*, vol. 65, no. 10, pp. 2517 – 2530, May 2017

J65. J. Tang, W. P. Tay, T. Q. S. Quek, and B. Liang, "System cost minimization in cloud RAN with limited fronthaul capacity," *IEEE Transactions on Wireless Communications*, vol. 16, no. 5, pp. 3371 – 3384, May 2017

J66. Y. Wang, W. P. Tay, and W. Hu, "A multitask diffusion strategy with optimized inter-cluster cooperation," *IEEE Journal of Selected Topics in Signal Processing*, vol. 11, no. 3, pp. 504 – 517, Mar. 2017

J67. M. Leng, W. P. Tay, F. Quitin, C. Cheng, S. G. Razul, and C. M. S. See, "Anchor-aided joint localization and synchronization using SOOP: Theory and experiments," *IEEE Transactions on Wireless Communications*, vol. 15, no. 11, pp. 7670 – 7685, Nov. 2016

J68. Y. Zhang, W. P. Tay, K. H. Li, M. Esseghir, and D. Gaiti, "Learning temporal-spatial spectrum reuse," *IEEE Transactions on Communications*, vol. 64, no. 7, pp. 3092 – 3103, Jul. 2016

J69. W. Luo, W. P. Tay, and M. Leng, "Infection spreading and source identification: A hide and seek game," *IEEE Transactions on Signal Processing*, vol. 64, no. 16, pp. 4228 – 4243, Aug. 2016

J70. J. Ho, W. P. Tay, T. Q. S. Quek, and E. K. P. Chong, "Robust decentralized detection and social learning in tandem networks," *IEEE Transactions on Signal Processing*, vol. 63, no. 19, pp. 5019 – 5032, Oct. 2015, **IEEE Signal Processing Society Young Author Best Paper Award**

J71. J. Tang, W. P. Tay, and T. Q. S. Quek, "Cross-layer resource allocation with elastic service scaling in cloud radio access network," *IEEE Transactions on Wireless Communications*, vol. 14, no. 9, pp. 5068 – 5081, Sep. 2015

J72. W. Hu and W. P. Tay, "Multi-hop diffusion LMS for energy-constrained distributed estimation," *IEEE Transactions on Signal Processing*, vol. 63, no. 15, pp. 4022 – 4036, Aug. 2015

J73. W. Xu, F. Quitin, M. Leng, W. P. Tay, and S. G. Razul, "Distributed localization of a RF target in NLOS environments," *IEEE Journal on Selected Areas in Communications*, vol. 33, no. 7, pp. 1 – 14, Jul. 2015

J74. W. P. Tay, "Whose opinion to follow in multihypothesis social learning? A large deviations perspective," *IEEE Journal of Selected Topics in Signal Processing*, vol. 9, no. 2, pp. 344 – 359, Mar. 2015

J75. M. Leng, W. P. Tay, T. Q. S. Quek, and H. Shin, "Distributed local linear parameter estimation using Gaussian SPAWN," *IEEE Transactions on Signal Processing*, vol. 63, no. 1, pp. 244 – 257, Jan. 2015

J76. Y. Zhang, W. P. Tay, K. H. Li, and D. Gaiti, "Distributed boundary estimation for spectrum sensing in cognitive radio networks," *IEEE Journal on Selected Areas in Communications*, vol. 32, no. 11, pp. 1961 – 1973, Nov. 2014

J77. W. Luo, W. P. Tay, and M. Leng, "How to identify an infection source with limited observations," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 4, pp. 586 – 597, Aug. 2014

J78. J. Tang, W. P. Tay, and Y. Wen, "Dynamic request redirection and elastic service scaling in cloud-centric media networks," *IEEE Transactions on Multimedia*, vol. 16, no. 5, pp. 1434 – 1445, Aug. 2014

J79. M. Leng, W. P. Tay, C. M. S. See, S. G. Razul, and M. Z. Win, "Modified CRLB for cooperative geolocation of two devices using signals of opportunity," *IEEE Transactions on Wireless Communications*, vol. 13, no. 7, pp. 3636 – 3649, Jul. 2014

J80. W. Hu and W. P. Tay, "An integer linear programming approach for a class of bilinear integer programs," *Operations Research Letters*, vol. 42, no. 3, pp. 226 —- 230, May 2014

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C113. Z. Madadi, F. Quitin, and W. P. Tay, "Periodic RF transmitter geolocation using a mobile receiver," in *Proc. IEEE International Conference Acoustics, Speech, and Signal Processing*, Brisbane, Australia, Apr. 2015

C114. Y. Zhang, W. P. Tay, K. H. Li, M. Esseghir, and D. Gaiti, "Distributed opportunistic spectrum access with spatial reuse in cognitive radio networks," in *Proc. IEEE Global Conference on Signal and Information Processing*, Atlanta, US, Dec. 2014

C115. J. Tang, W. P. Tay, and T. Q. S. Quek, "Cross-layer resource allocation in cloud radio access network," in *Proc. IEEE Global Conference on Signal and Information Processing*, Atlanta, US, Dec. 2014, **invited paper**

C116. W. Xu, F. Quitin, M. Leng, W. P. Tay, and S. G. Razul, "Distributed localization of a non-cooperative RF target in NLOS environments," in *Proc. International Conference on Information Fusion*, Salamanca, Spain, Jul. 2014

C117. W. Hu and W. P. Tay, "Generalized diffusion adaptation for energy-constrained distributed estimation," in *Proc. International Conference on Information Fusion*, Salamanca, Spain, Jul. 2014

C118. J. Ho, W. P. Tay, and T. Q. S. Quek, "Robust detection and social learning in tandem networks," in *Proc. IEEE International Conference Acoustics, Speech, and Signal Processing*, Florence, Italy, May 2014

C119. W. Luo and W. P. Tay, "Estimating infection sources in a network with incomplete observations," in *Proc. IEEE Global Conference on Signal and Information Processing*, Austin, USA, Dec. 2013

C120. W. Xu, M. Leng, W. P. Tay, and S. G. Razul, "Distributed localization of an unknown target in NLOS environments," in *Proc. International Conference on Information, Communications and Signal Processing*, Tainan, Taiwan, Dec. 2013

C121. W. Luo and W. P. Tay, "Finding an infection source under the SIS model," in *Proc. IEEE International Conference Acoustics, Speech, and Signal Processing*, Vancouver, Canada, May 2013

C122. Y. Zhang, W. P. Tay, K. H. Li, and D. Gaiti, "Distributed boundary estimation for spectrum sensing in cognitive radio networks," in *Proc. IEEE Wireless Communications and Networking Conference*, Shanghai, China, Apr. 2013

C123. M. Leng, W. P. Tay, C. M. S. See, and S. G. Razul, "Fundamental limits for location and velocity estimation using asynchronous beacons," in *Proc. IEEE Wireless Communications and Networking Conference*, Shanghai, China, Apr. 2013

C124. M. Leng, W. P. Tay, C. M. S. See, and S. G. Razul, "GPS-free localization using asynchronous beacons," in *Proc. IEEE Int. Conf. on Mobile Ad-hoc and Sensor Networks*, Chengdu, China, Dec. 2012, **invited paper**

C125. W. Luo and W. P. Tay, "Identifying multiple infection sources in a network," in *Proc. Asilomar Conference on Signals, Systems and Computers*, Asilomar, USA, Nov. 2012, **invited paper, best student paper award**

C126. Y. Nijsure, W. P. Tay, E. Gunawan, and J. Lai, "A Bayesian nonparametric approach to tumor detection using UWB imaging," in *Proc. of IEEE International Conference on Ubiquitous Wireless Broadband*, New York, USA, Sep. 2012

C127. F. Wen and W. P. Tay, "Localization for mixed near-field and far-field sources using data supported optimization," in *Proc. International Conference on Information Fusion*, Singapore, Jul. 2012

C128. F. Wen and W. P. Tay, "Tensor decomposition based R-dimensional matrix pencil method," in *Proc. International Conference on Information Fusion*, Singapore, Jul. 2012

C129. W. Luo and W. P. Tay, "Identifying infection sources in large tree networks," in *Proc. IEEE International Conference on Sensing, Communication, and Networking*, Seoul, Korea, Jun. 2012

C130. C. Cheng, W. P. Tay, and G. B. Huang, "Extreme learning machines for intrusion detection," in *Proc. IEEE Int. Joint Conf. on Neural Networks*, Brisbane, Australia, Jun. 2012

C131. M. Leng, W. P. Tay, and T. Q. S. Quek, "Cooperative and distributed localization for wireless sensor networks in multipath environments," in *Proc. IEEE International Conference Acoustics, Speech, and Signal Processing*, Kyoto, Japan, Mar. 2012

C132. M. Leng, W. P. Tay, and T. Q. S. Quek, "Cooperative and distributed localization for wireless sensor networks in multipath environments," in *Proc. International Conference on Information, Communications and Signal Processing*, Singapore, Dec. 2011

C133. D. W. Soh, T. Q. S. Quek, and W. P. Tay, "Randomized rumor spreading in non-static networks," in *Proc. IEEE Int. Conf. on ICT Convergence*, Seoul, Korea, Sep. 2011, **invited paper**

C134. D. W. Soh, T. Q. S. Quek, and W. P. Tay, "Randomized broadcast in dynamic network environments," in *Proc. IEEE Workshop on Signal Processing Advances in Wireless Communications*, San Francisco, USA, Jun. 2011

C135. W. P. Tay and J. N. Tsitsiklis, "Error exponents for decentralized detection in feedback architectures," in *Proc. IEEE International Conference Acoustics, Speech, and Signal Processing*, Prague, Czech Republic, May 2011

C136. W. P. Tay and J. N. Tsitsiklis, "The value of feedback for decentralized detection in large sensor networks," in *Proc. Int. Sym. on Wireless and Pervasive Computing*, Hong Kong, Feb. 2011

C137. W. P. Tay, J. N. Tsitsiklis, and M. Z. Win, "On the sub-exponential decay of detection probabilities in long tandems," in *Proc. IEEE International Conference Acoustics, Speech, and Signal Processing*, Honolulu, USA, Apr. 2007, pp. 837–840

C138. W. P. Tay, J. N. Tsitsiklis, and M. Z. Win, "Bayesian detection in bounded height tree networks," in *Proc. of Data Compression Conference*, Snowbird, USA, Mar. 2007, pp. 243–252

C139. W. P. Tay, J. N. Tsitsiklis, and M. Z. Win, "Detection in dense wireless sensor networks," in *Proc. IEEE Wireless Communications and Networking Conference*, Hong Kong, Mar. 2007, pp. 3483–3488

C140. W. P. Tay, J. N. Tsitsiklis, and M. Z. Win, "Data fusion trees for detection: Does architecture matter?" in *Proc. Allerton Conference on Communications, Control and Computing*, Monticello, USA, Sep. 2006

C141. W. P. Tay, J. N. Tsitsiklis, and M. Z. Win, "Asymptotically optimal distributed censoring," in *Proc. IEEE International Symposium on Information Theory*, Seattle, USA, Jul. 2006, pp. 625–629

C142. W. P. Tay, J. N. Tsitsiklis, and M. Z. Win, "Censoring sensors: Asymptotics and the value of cooperation," in *Proc. Conference on Information Science and Systems*, Princeton, USA, Mar. 2006, pp. 62–67

2.5 Book Chapters and Monographs

B1. X. Jian, F. Ji, and W. P. Tay, "Generalizing graph signal processing: High dimensional spaces, models and structures," *Foundations and Trends® in Signal Processing*, vol. 17, no. 3, pp. 209–290, 2023. [Online]. Available: `http://dx.doi.org/10.1561/2000000119`

B2. F. Ji, W. Tang, J. Yang, and W. P. Tay, "Online information spreading and source identification," in *Online Social Networks: Perspectives, Applications and Developments*, C. W. Tan, Ed. Hauppauge, NY: Nova Science Publishers, Inc., 2020, ch. 4

B3. R. Rabiee, I. Bajaj, and W. P. Tay, "Vehicle localization in GNSS-denied environments," in *Cooperative Localization and Navigation: Theory, Research, and Practice*, C. Gao, G. Zhao, and H. Fourati, Eds. Boca Raton, FL: CRC Press, 2019, ch. 11, pp. 199 – 222

B4. M. Leng and W. P. Tay, "Fundamental limits of self-localization for cooperative robotic platforms using signals of opportunity," in *Cooperative Robots and Sensor Networks 2015*, ser. Studies in Computational Intelligence, A. Koubâa and J. R. M. de Dios, Eds. Cham: Springer, 2015, vol. 604, pp. 159 – 181

B5. W. P. Tay and J. N. Tsitsiklis, "Error exponents for decentralized detection in tree networks," in *Networked Sensing Information and Control*, V. Saligrama, Ed. Boston, MA: Springer, 2008, pp. 73 – 92

2.6 Thesis

1. W. P. Tay, "Decentralized detection in resource-limited sensor network architectures," Ph.D. dissertation, Massachusetts Institute of Technology, Dec. 2007

2.7 Other Publications

O1. Z. Madadi, F. Quitin, and W. P. Tay, "Vehicle localization using periodic transmissions from an RSU in GNSS denied environments," in *Proc. ITS Asia-Pacific Forum*, Nanjing, China, Apr. 2015

O2. Y. Nijsure, W. P. Tay, and E. Gunawan, "An impulse radio ultra wideband system for contactless non-invasive respiratory monitoring," *Asia Pacific BioTech News*, vol. 16, no. 10, pp. 28–31, Oct. 2012 , ***invited paper***

2.8 Patents and Disclosures

2.8.1 Patents Filed

PF1. S. Wang, R. She, Q. Kang, T. Geng, and W. P. Tay, "System and method for processing image data," Singapore PCT SG2025/050 428, Jun. 25, 2025

PF2. C. Jin, W. P. Tay, K. Zhao, K. V. Ling, and J. Lu, "Device, positioning system and method of determining a position of the device for vehicular communication," Singapore Patent Application 11 202 501 107V, Sep. 19, 2023

PF3. Q. Kang, R. She, W. P. Tay, N. D. Navarro, R. Khurana, A. Hartmannsgruber, and Y. L. Guan, "Position determination of a vehicle using image segmentations," PCT EP2023/063 085, May 16, 2023

PF4. R. She, Q. Kang, S. Wang, W. P. Tay, N. D. Navarro, and A. Hartmannsgruber, "Computer-implemented method for estimating a position and/or pose of a vehicle," PCT EP2024/061 019, Apr. 23, 2024

PF5. S. Wang, Q. Kang, R. She, W. P. Tay, N. D. Navarro, and A. Hartmannsgruber, "A computer-implemented method for camera pose regression in a challenging traffic environment," UK Patent Application 202 216 510, Nov. 7, 2022

PF6. Q. Kang, R. She, W. P. Tay, R. Khuarana, N. D. Navarro, S. Wang, and A. Hartmannsgruber, "Method for image segmentation matching," PCT EP2023/063 041, May 17, 2022

PF7. F. Ji, Pratibha, and W. P. Tay, "Methods and systems for recovering folded signals," PCT SG2020/050 631, May 14, 2021

PF8. C. Jin, I. Bajaj, K. Zhao, W. P. Tay, and K. V. Ling, "5G positioning using code phase and carrier phase receiver," PCT SG2021/050 732, Nov. 27, 2020

2.8.2 Patents Granted

PG1. Y. Wang, W. P. Tay, J. S. Kee, and K. Thangamariappan, "Wireless sensor network, parameter optimization method thereof and warehouse system," China Patent 111 148 109, May 12, 2020

PG2. Y. Wang, W. P. Tay, J. S. Kee, and K. Thangamariappan, "Wireless sensor network and parameter optimization method thereof, and warehouse system," U.S. Patent 20 200 146 103, May 7, 2020

2.8.3 Technology Disclosures

TD1. X. Ran, W. P. Tay, and C. H. T. Lee, "A deep reinforcement learning-based novel false data injection attack detection in power systems," Singapore Technology Disclosure NTU Ref: 2022-411, Nov. 9, 2022

TD2. W. Wang, M. Yang, Y. L. Guan, and W. P. Tay, "Intelligent transportation system runtime middleware," Singapore Technology Disclosure NTU Ref: 2021-269, May 31, 2021

TD3. C. Wang, Q. Zhang, and W. P. Tay, "Energy efficient decentralized real-time multi-camera multi-target recognition and tracking system," Singapore Technology Disclosure NTU Ref: 2020-388, Dec. 1, 2020

TD4. C. Yang, M. Yang, Y. L. Guan, and W. P. Tay, "Intelligent transportation system use-case software package," Singapore Technology Disclosure NTU Ref: 2020-028, Feb. 6, 2020

TD5. W. P. Tay, X. Zhong, R. Rabiee, and Y. Yan, "Lane-level vehicle tracker for V2X communication systems," NTU Copyright TD/206/17, Oct. 12, 2017

TD6. X. He, W. P. Tay, and M. Sun, "Privacy engine for IoT device networks," NTU Copyright TD/071/16, May 26, 2016

TD7. W. P. Tay, F. Quitin, and Z. Madadi, "Geolocation using virtual TDOA with asynchronous clocks," NTU Copyright TD/249/14, Jan. 9, 2015

TD8. W. P. Tay and F. Quitin, "Localization of non-cooperative RF targets in cluttered environments," NTU Copyright TD/078/14, Jun. 24, 2014

TD9. W. Hu and W. P. Tay, "A generalized diffusion adaptation strategy for energy-constrained estimation," NTU Copyright TD/077/14, May 30, 2014

TD10. W. P. Tay, E. Gunawan, and Y. L. Guan, "Method for contactless respiratory monitoring using multiple UWB transceivers," NTU Copyright TD/215/12, Feb. 26, 2012

2.9 Selected Invited Presentations and Keynotes

Keynote and Distinguished Presentations

1. Keynote presentation, "Continuous Dynamics in Graph Neural Networks," Workshop on Graphs and other Complex Structures at AAAI 2026, Jan. 2026
2. Keynote speaker, "Data-driven Inference Privacy Sanitization," International Conference on Computer Technology and Information Science, Wuhan, China, Jun. 2023
3. Plenary speaker at the Graph Signal Processing Workshop, Madrid, Spain in 2020. Unable to attend due to travel uncertainty due to COVID-19 restrictions.
4. Keynote speaker (distinguished presentation), "Generalized graph signal processing," 7th IEEE Global Conference on Signal and Information Processing (GlobalSIP), Ottawa, Canada, Nov. 2019

Invited Talks and Presentations

1. Invited talk, "Continuous Dynamics in Graph Neural Networks," Tsinghua University, Beijing, China, Jul. 2024

2. Invited presentation, "Robust graph neural diffusion for image matching," IEEE International Conference on Image Processing, Kuala Lumpur, Malaysia, Oct. 2023
3. Invited tutorial, "Advanced graph signal processing," IEEE International Conference on Acoustics, Speech and Signal Processing, May 2022
4. Invited talk, "Data-driven privacy sanitization with regularization," Kookmin University, South Korea, Feb. 2022
5. Invited tutorial, "Privacy for IoT: Signal processing theories and methods," 7th IEEE World Forum on the Internet of Things, New Orleans, USA, Jun. 2021
6. Invited talk, "COSMO: Next gen V2X architecture & ecosystem for smart mobility," Advanced Manufacturing and Engineering Day, Feb. 2021
7. Invited presentation, "Fake News Detection and Source Identification," DSO National Laboratories, Oct. 2019
8. Invited presentation, "Inference and data privacy in IoT networks," IEEE Workshop on Signal Processing Advances in Wireless Commun., Hokkaido, Japan, Jul. 2017
9. Invited talk, "Towards Information Privacy for the Internet of Things," Singapore University of Technology and Design, Singapore, Oct. 2016
10. Invited talk, "Algorithms for Network Infection Sources Estimation," Singapore University of Technology and Design, Singapore, Jun. 2016
11. Invited presentation, "Posterior Cramér-Rao lower bound for mobile emitter tracking based on a TDOA-FDOA multi-measurement model," IEEE Int. Conf. on Ubiquitous Wireless Broadband, Nanjing, China, Oct. 2016
12. Invited presentation, "TDOA-FDOA multiple target detection and tracking in the presence of measurement errors and biases," IEEE Workshop on Signal Proc. Advances in Wireless Commun., Edinburgh, UK, Jul. 2016
13. Invited presentation, "Joint resource segmentation and transmission rate adaptation in cloud RAN with caching as a service," IEEE Workshop on Signal Proc. Advances in Wireless Commun., Edinburgh, UK, Jul. 2016
14. Invited talk, "Algorithms for network infection sources estimation," Chongqing University, Jun. 2016
15. Invited talk, "Proof of concept study using DSRC, IMU and map fusion for vehicle localization in GNSS-denied environments," Chongqing University of Posts and Telecommunications, Jun. 2016
16. Invited talk, "Towards information privacy for the Internet of Things," International Conference on Communications and Networking in China (Chinacom), Chonqqing, China, Jun. 2016
17. Invited presentation, "Towards system cost minimization in cloud radio access network," Asilomar Conf. on Signals, Systems and Computers, Asilomar, USA, Nov. 2015
18. Tan Chin Tuan Fellow seminar, "Network infection sources estimation," University of Illinois at Urbana-Champaign, USA, Jul. 2015
19. Tan Chin Tuan Fellow seminar, "Asymptotics in social learning", University of Illinois at Urbana-Champaign, USA, Jul. 2015
20. Invited talk, "Collaborative Social Sensing in IoT," KTH-NTU workshop on Urban Social Systems and Networking, NTU, Nov. 2015
21. Invited talk, "Social Learning in IoT Under Uncertainty," Cybersecurity Workshop with Tel Aviv University, NTU, Oct. 2015
22. Invited talk, "Identifying Infection Sources in a Network," Singapore-UK Cybersecurity Workshop, NTU, Feb. 2015

23. Invited talk, "iMata: Information Management in Twitter Rumor Attacks," Singapore-UK Cybersecurity Workshop, NTU, Feb. 2015
24. Invited talk, "Identifying Infection Sources in a Network," National University of Singapore, Singapore, Jan. 2015
25. Invited talk, "Asymptotics in Social Learning," Hong Kong University, Jan. 2015
26. Invited presentation, "Joint scheduling and localization in UWB networks," IEEE Int. Conf. on Commun., London, UK, Jun. 2015
27. Invited presentation, "Whose opinion to follow in social learning?", IEEE Taiwan-Hong Kong Joint Workshop on Information Theory and Communications, Hong Kong, Jan. 2015
28. Invited presentation, "Cross-layer resource allocation in cloud radio access network," IEEE Global Conf. on Signal and Information Processing, Atlanta, US, Dec. 2014
29. Invited talk, "Advanced Communications and Security for Future Large-scale Vehicle-to-vehicle Networks," NTU-KTH Workshop, NTU, Sep. 2014
30. Invited talk, "Network Infection Source Estimation," University of Technology of Troyes, France, Jul. 2014
31. Invited talk, "Distributed Localization of a Non-cooperative RF Target in NLOS Environments," Positioning, Navigation and Timing Workshop, Haiwaii, USA, Jun. 2014
32. Invited talk, "Fundamental Limits for Geolocation Using Signals of Opportunity," Positioning, Navigation and Timing Workshop, Haiwaii, USA, Jun. 2014
33. Invited presentation, "GPS-free localization using asynchronous beacons," IEEE Int. Conf. on Mobile Ad-hoc and Sensor Networks, Chengdu, China, Dec. 2012
34. Invited presentation, "Identifying multiple infection sources in a network," Asilomar Conf. on Signals, Systems and Computers, Asilomar, USA, Nov. 2012
35. Invited talk, "Distributed Detection and Tracking using Wireless Sensor Networks," Maritime and Port Authority of Singapore, Singapore, Feb. 2012
36. Invited presentation, "Randomized rumor spreading in non-static networks," IEEE Int. Conf. on ICT Convergence, Seoul, Korea, Sep. 2011, invited presentation
37. Invited talk, "Distributed Decision Making," Centre for Strategic Infocomm Technologies, Singapore, Jun. 2010
38. Invited talk, "Information Fusion in Large Networks," Land Transport Authority, Singapore, May 2010
39. Invited talk, "Distributed Decision Making in Tree Networks," NTU-UTT Workshop, NTU, Apr. 2010
40. Invited talk, "Decentralized Detection in Large Sensor Networks," Qualcomm Inc., USA, Mar. 2007

3 Educational Activities

3.1 Completed Ph.D. Dissertation Supervisions

1. Luo Wuqiong, 08/2010 – 08/2014, "Identifying infection sources in a network."
2. Tang Jianhua, 01/2011 – 01/2015, "Elastic service scaling optimization in cloud-based communication systems."
3. Muhammad Sibtain Hamayun, 08/2010 – 08/2015, "Exploiting statistical side information to optimize secondary spectrum access."

4. Zhang Yi, 01/2011 – 01/2015, "Learning methods for temporal-spatial opportunistic spectrum access in cognitive radio networks."
5. Sun Meng, 08/2014 – 08/2018, "Privacy-preserving decentralized detection in sensor networks."
6. Wang Yuan, 08/2012 – 08/2018, "Cooperative inference and learning for Internet-of-Things with limited resources."
7. Ho Jun Feng Jack, 08/2012 – 11/2018, "Learning models in social networks."
8. Yang Jielong, 08/2015 – 08/2019, "On truth finding in multi-agent networks."
9. Tang Wenchang, 08/2015 – 08/2019, "Identifying misinformation and their sources in social networks."
10. Kang Qiyu, 08/2015 – 08/2019, "Sequential crowdsourcing and recommendation strategies."
11. Lau Tze Siong, 08/2015 – 01/2020, "Operationally constrained quickest change detection with multiple post-change distributions." Commendation for EEE Doctorate Research Excellence Award.
12. Wang Chongxiao, 08/2016 – 12/2021, "Inference privacy preservation in linear and computational systems."

3.2 Current Ph.D. Dissertation Supervisions

1. Lu Yiqi, 08/2018 – present, "Machine learning for inference in graphs."
2. Lee See Hian, 08/2019 – present, "E-commerce knowledge graphs."
3. Vivek Mohan, 01/2020 – present, "Unconventional sampling algorithms and hardware for brain-machine interfaces."
4. Jian Xingchao, 08/2020 – present, "Processing stochastic time-varying signals over graphs."
5. Zhao Kai, 01/2021 – present, "Graph neural networks for multi-sensor pattern recognition."
6. Wang Sijie, 08/2021 – present, "Segmentation in autonomous driving environment."
7. Zhang Purui, 08/2022 – present, "High-order graph signal processing and neural networks."
8. Li Disheng, 01/2023 – present, "Machine learning methods for urban localization."
9. Liang Wenfei, 01/2023 – present, "Federated learning for graph neural networks."
10. Zhao Yanan, 01/2023 – present, "Generalized graph signal processing over simplicial complexes."

3.3 Completed M.Eng. Dissertation Supervisions

1. Cheng Chi, 08/2011 – 05/2015, "Exploring the use of signals-of-opportunity for practical localization."

3.4 Courses Taught

- EE7401, Probability and Random Processes (about 70 students per course).
- IE4497, Pattern Recognition and Deep Learning (about 25 students per course).
- IE0005, Introduction to Data Science and Artificial Intelligence (about 500 students per course).
- EE2008/IM1001, Data Structures and Algorithms (about 300 students per course).
- EE4105/IM4105, Cellular Communication System Design (about 110 students per course).
- EE6713, Network Design and Simulation (about 40 students per course).
- EE7101, Fundamentals of Information Theory (about 10 students per course).

3.5 Courses Developed

- 2012, proposed and developed new course EE7101 – Fundamentals of Information Theory.
- 2015, developed new course materials for EE2008/IM1001 – Data Structures and Algorithms due to increase of weekly lecture hours from 2 hours to 3 hours.
- 2015, developed Technology Enabled Learning video lectures and online learning activities for EE2008/IM1001 – Data Structures and Algorithms.
- 2015, coordinated and developed online assessment and practice (OASIS) for EE2008/IM1001 – Data Structures and Algorithms.
- 2021, revamped and developed new course materials for EE4497 Pattern Recognition and Machine Learning.

4 Professional Activities

4.1 University/National Committee Activities

- NTU Teaching Council, Aug. 2026 – present.
- Associate Chair (Faculty), School of Electrical and Electronic Engineering, Jul. 2025 – present.
- Member, School Review Committee for Promotion and Tenure, School of Art, Design and Media, Jul. 2024 – present.
- Member, Technical Committee on Intelligent Transport System, Enterprise Singapore, for ITS standards development, Jul. 2022 – Dec. 2024.
- Member, Academic Institutions Curriculum Advisory Committee, Singapore 5G & Telecoms Academy, Jun. 2021 – Dec. 2023.
- Associate Chair (Academic), School of Electrical and Electronic Engineering, Feb. 2020 – Jun. 2025.
- Assistant Chair (Academic), School of Electrical and Electronic Engineering, Jun. 2017 – Jan. 2020.
- Member, NTU Presidential Postdoctoral Fellowship 2023 Evaluation Committee, Jun. 2023.
- Cluster Director, Cyber Physical System for Critical Information Infra-structure Research Program, Energy Research Institute, Apr. 2019 – 31 Mar. 2025.
- Program Director, Advanced Sensing Technologies, Centre for Information Sciences and Systems, Jan. 2020 – present.
- Program Director, Cyber and Network Security, INFINITUS, Centre for Infocomm Technology, Jan. 2015 – Sep. 2016.
- Program Director, Communications and Network Systems, INFINITUS, Centre for Infocomm Technology, Jan. 2013 – Dec. 2014.
- Member, School of EEE outreach committee Jan. 2013 – Dec. 2013.

4.2 Professional Society Activities

- Senior Area Editor, IEEE Transactions on Signal Processing, 2026 – present.
- Academy Mentor, EURASIP Academy, 2024 – present.
- Subject Editor, Signal Processing, Elsevier, 2024 – present.
- Associate Editor, IEEE Internet of Things Journal, 2023 – present.

- Associate Member, Signal Processing Theory and Methods Technical Committee, IEEE Signal Processing Society, 2023 – present.
- Associate Editor, IEEE Transactions on Signal and Information Processing over Networks, 2019 – 2024.
- Editor, IEEE Open Journal of Vehicular Technology, 2019 – 2023.
- Editor, IEEE Transactions on Wireless Communications, 2017 – 2023.
- Associate Editor, IEEE Transactions on Signal Processing, 2015 – 2019.
- Guest Editor, IEEE Transactions on Signal and Information Processing over Networks Special Issue on Distributed Information Processing in Social Networks, 2016 – 2017.
- Member, Machine Learning for Signal Processing Technical Committee, IEEE Signal Processing Society, 2015 – 2020.
- Member, Internet of Things Special Interest Group, IEEE Signal Processing Society, 2015 – 2019.
- Chair, Interest Group on Distributed and Sensor Networks for Mobile Media Computing and Applications, IEEE Communications Society Technical Committee on Multimedia Communications, 2014.
- Vice-Chair, Interest Group on Green Multimedia Communications, IEEE Communications Society Technical Committee on Multimedia Communications, 2012 – 2014.

4.3 Conference Organizing Committees

- Advisory Chair, International Conference on Embodied Intelligence and Robotics (EIR), 2026.
- Tutorial Chair, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2026.
- Program Chair, International Conference on Signal Processing and Information Communications (ICSPIC), 2023.
- Signal Processing Grand Challenges Co-chair, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2022.
- Web Chair, IEEE Global Communications Conference (Globecom), 2017.
- Tutorial Chair, International Conference on Information, Communications and Signal Processing, 2015.
- Organizer of special session on "Signal processing for social networks" in IEEE Conference on Digital Signal Processing, 2015.
- Publicity Chair, International Conference on Information, Communications and Signal Processing, 2013.
- Organizer of special session on "Emerging technologies in cooperative communication networks" in IEEE International Conference on Information, Communications and Signal Processing, 2011.
- Session chair
  - IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2015, 2016, 2017, 2018, 2019, 2020, 2022
  - IEEE Global Conference on Signal and Information Processing (GlobalSIP) 2018, 2019

4.4 Technical Program Committees

- Conference on Computer Vision and Pattern Recognition (CVPR) 2025 – present
- European Conference on Computer Vision (ECCV) 2025 – present
- International Conference on Machine Learning (ICML) 2024 – present

- International Conference on Learning Representations (ICLR) 2024 – present
- Neural Information Processing Systems (NeurIPS) 2023 – present
- International Joint Conference on Artificial Intelligence (IJCAI) 2022 – present
- AAAI Conference on Artificial Intelligence (AAAI) 2021 – present
- European Signal Processing Conference (EUSIPCO) 2017 – present
- IEEE Global Conference on Signal and Information Processing (GlobalSIP) 2017, 2018, 2019
- IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2016 – present
- IEEE International Workshop on Machine Learning for Signal Processing (MLSP) 2015 – present